```python
import pandas as  pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("311_Service_Requests.csv", low_memory=False)

df.head()
```

```
   Unique Key            Created Date              Closed Date
Agency  \
0    32310363  12/31/2015 11:59:45 PM  01/01/2016 12:55:15 AM   NYPD

1    32309934  12/31/2015 11:59:44 PM  01/01/2016 01:26:57 AM   NYPD

2    32309159  12/31/2015 11:59:29 PM  01/01/2016 04:51:03 AM   NYPD

3    32305098  12/31/2015 11:57:46 PM  01/01/2016 07:43:13 AM   NYPD

4    32306529  12/31/2015 11:56:58 PM  01/01/2016 03:24:42 AM   NYPD


                        Agency Name           Complaint Type  \
0  New York City Police Department  Noise - Street/Sidewalk
1  New York City Police Department         Blocked Driveway
2  New York City Police Department         Blocked Driveway
3  New York City Police Department          Illegal Parking
4  New York City Police Department          Illegal Parking

                      Descriptor    Location Type  Incident Zip  \
0              Loud Music/Party  Street/Sidewalk       10034.0
1                     No Access  Street/Sidewalk       11105.0
2                     No Access  Street/Sidewalk       10458.0
3  Commercial Overnight Parking  Street/Sidewalk       10461.0
4              Blocked Sidewalk  Street/Sidewalk       11373.0

        Incident Address  ... Bridge Highway Name Bridge Highway
Direction  \
0    71 VERMILYEA AVENUE  ...                  NaN
NaN
1        27-07 23 AVENUE  ...                  NaN
NaN
2  2897 VALENTINE AVENUE  ...                  NaN
NaN
3    2940 BAISLEY AVENUE  ...                  NaN
NaN
4          87-14 57 ROAD  ...                  NaN
NaN

  Road Ramp Bridge Highway Segment Garage Lot Name Ferry Direction  \
0       NaN                     NaN              NaN              NaN
```

```
1       NaN                     NaN             NaN             NaN
2       NaN                     NaN             NaN             NaN
3       NaN                     NaN             NaN             NaN
4       NaN                     NaN             NaN             NaN

  Ferry Terminal Name   Latitude  Longitude  \
0                 NaN  40.865682 -73.923501
1                 NaN  40.775945 -73.915094
2                 NaN  40.870325 -73.888525
3                 NaN  40.835994 -73.828379
4                 NaN  40.733060 -73.874170

                                    Location
0   (40.86568153633767, -73.92350095571744)
1  (40.775945312321085, -73.91509393898605)
2  (40.870324522111424, -73.88852464418646)
3   (40.83599404683083, -73.82837939584206)
4  (40.733059618956815, -73.87416975810375)

[5 rows x 53 columns]
```

```
list(df.columns)
```

```
['Unique Key',
 'Created Date',
 'Closed Date',
 'Agency',
 'Agency Name',
 'Complaint Type',
 'Descriptor',
 'Location Type',
 'Incident Zip',
 'Incident Address',
 'Street Name',
 'Cross Street 1',
 'Cross Street 2',
 'Intersection Street 1',
 'Intersection Street 2',
 'Address Type',
 'City',
 'Landmark',
 'Facility Type',
 'Status',
 'Due Date',
 'Resolution Description',
 'Resolution Action Updated Date',
 'Community Board',
 'Borough',
 'X Coordinate (State Plane)',
 'Y Coordinate (State Plane)',
```

```
 'Park Facility Name',
 'Park Borough',
 'School Name',
 'School Number',
 'School Region',
 'School Code',
 'School Phone Number',
 'School Address',
 'School City',
 'School State',
 'School Zip',
 'School Not Found',
 'School or Citywide Complaint',
 'Vehicle Type',
 'Taxi Company Borough',
 'Taxi Pick Up Location',
 'Bridge Highway Name',
 'Bridge Highway Direction',
 'Road Ramp',
 'Bridge Highway Segment',
 'Garage Lot Name',
 'Ferry Direction',
 'Ferry Terminal Name',
 'Latitude',
 'Longitude',
 'Location']
```

```
df.shape
```

```
(364558, 53)
```

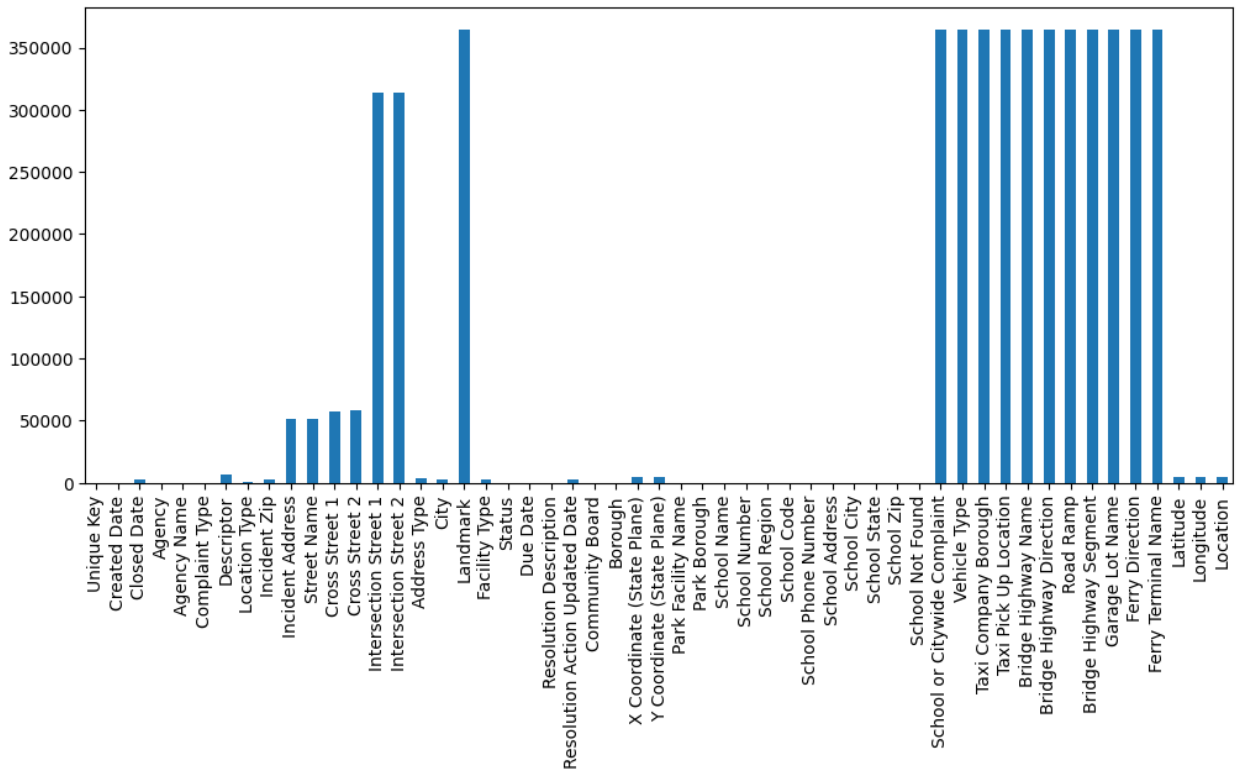
```
df.isnull().sum()
```

```
Unique Key                              0
Created Date                            0
Closed Date                          2381
Agency                                  0
Agency Name                             0
Complaint Type                          0
Descriptor                           6501
Location Type                         133
Incident Zip                         2998
Incident Address                    51699
Street Name                         51699
Cross Street 1                      57188
Cross Street 2                      57805
Intersection Street 1              313438
Intersection Street 2              314046
Address Type                         3252
City                                 2997
```

```
Landmark                            364183
Facility Type                         2389
Status                                   0
Due Date                                 3
Resolution Description                   0
Resolution Action Updated Date        2402
Community Board                          0
Borough                                  0
X Coordinate (State Plane)            4030
Y Coordinate (State Plane)            4030
Park Facility Name                       0
Park Borough                             0
School Name                              0
School Number                            0
School Region                            1
School Code                              1
School Phone Number                      0
School Address                           0
School City                              0
School State                             0
School Zip                               1
School Not Found                         0
School or Citywide Complaint        364558
Vehicle Type                        364558
Taxi Company Borough                364558
Taxi Pick Up Location               364558
Bridge Highway Name                 364261
Bridge Highway Direction            364261
Road Ramp                           364296
Bridge Highway Segment              364296
Garage Lot Name                     364558
Ferry Direction                     364557
Ferry Terminal Name                 364556
Latitude                              4030
Longitude                             4030
Location                              4030
dtype: int64
```

```
#Draw a frequency plot to show the number of null values in each
column of the DataFrame
df.isnull().sum().plot(kind='bar',figsize=(12,5))
```

```
<Axes: >
```




```
#finding out empty columns
empty_cols  = df.columns[df.isnull().all()]
empty_cols
```

```
Index(['School or Citywide Complaint', 'Vehicle Type', 'Taxi Company
Borough',
       'Taxi Pick Up Location', 'Garage Lot Name'],
      dtype='object')
```

```
#Dropping Empty Columns
df = df.drop(['School or Citywide Complaint', 'Vehicle Type', 'Taxi
Company Borough',
       'Taxi Pick Up Location', 'Garage Lot Name'],axis=1)
```

```
df.shape
```

```
(364558, 48)
```

```
#Remove the records whose Closed Date values are null
df = df.dropna(subset=['Closed Date'])
```

```
df['Created Date'] = pd.to_datetime(df['Created Date'],
errors='coerce')
```

```
C:\Users\surya\AppData\Local\Temp\ipykernel_7732\1175343188.py:1:
UserWarning: Could not infer format, so each element will be parsed
individually, falling back to `dateutil`. To ensure parsing is
```

```
consistent and as-expected, please specify a format.
  df['Created Date'] = pd.to_datetime(df['Created Date'],
errors='coerce')
```

```
df['Closed Date'] = pd.to_datetime(df['Closed Date'], errors='coerce')
```

```
C:\Users\surya\AppData\Local\Temp\ipykernel_7732\1584812765.py:1:
UserWarning: Could not infer format, so each element will be parsed
individually, falling back to `dateutil`. To ensure parsing is
consistent and as-expected, please specify a format.
  df['Closed Date'] = pd.to_datetime(df['Closed Date'],
errors='coerce')
```

```
#Analyze the date column, and remove entries that have an incorrect
timeline
df = df[df['Closed Date'] >= df['Created Date']]
```

```
#Calculate the time elapsed in closed and creation date
df['Request_Closing_Time'] =  df['Closed Date'] - df['Created Date']
```

```
#Convert the calculated date to seconds to get a better representation
df['Request_Closing_Time'] =
df['Request_Closing_Time'].dt.total_seconds()
```

```
#View the descriptive statistics for the newly created column
df['Request_Closing_Time'].describe()
```

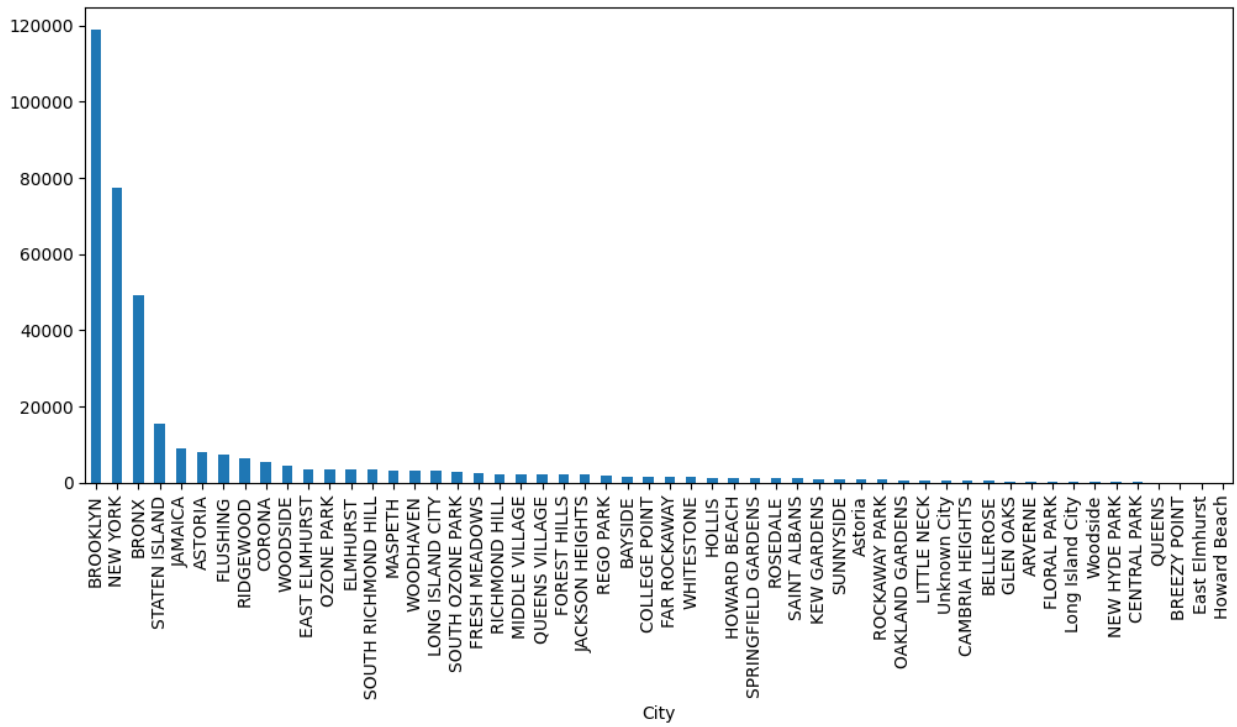
```
count    3.621770e+05
mean     1.511330e+04
std      2.110255e+04
min      6.100000e+01
25%      4.533000e+03
50%      9.616000e+03
75%      1.887800e+04
max      2.134342e+06
Name: Request_Closing_Time, dtype: float64
```

```
#Check the number of null values in the Complaint_Type and City
columns
df['City'].isnull().sum()
```

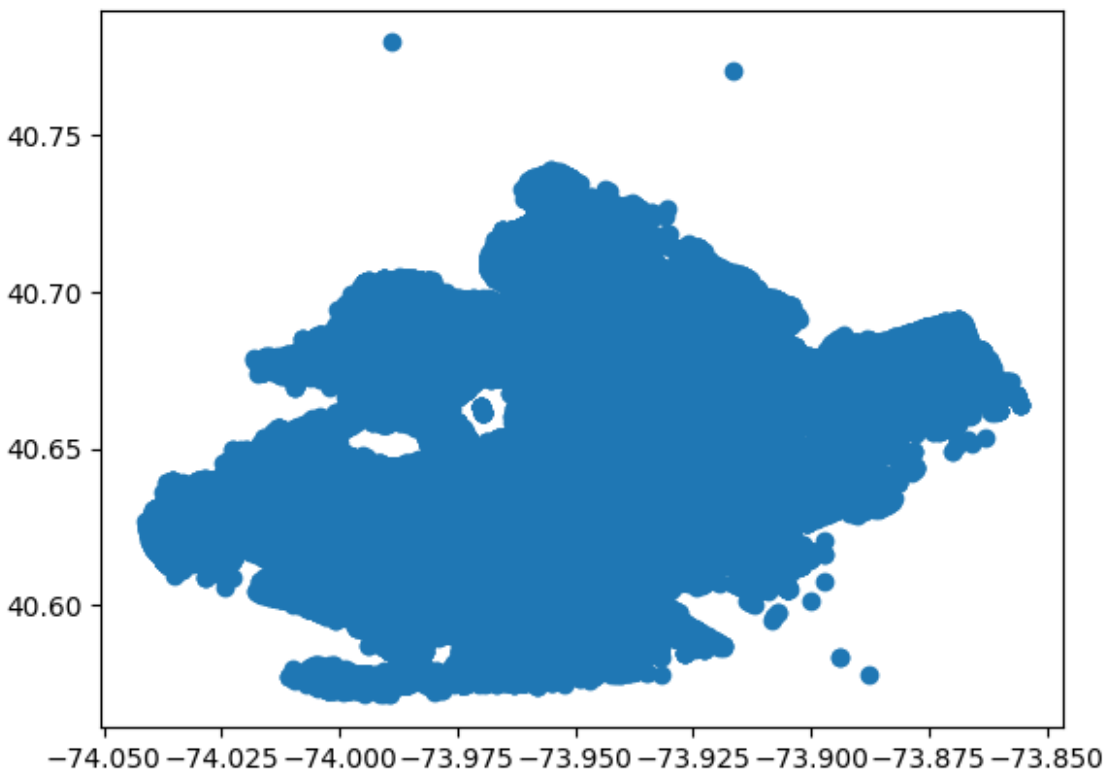
```
np.int64(674)
```

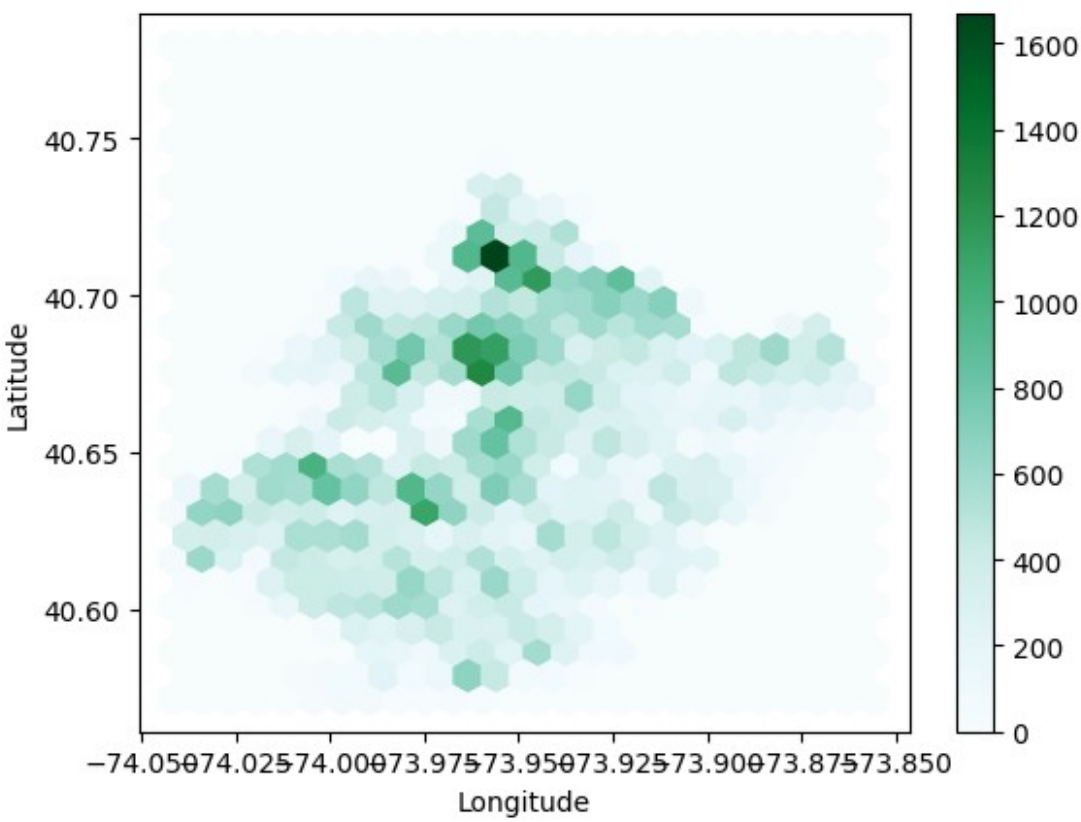
```
#Impute the NA value with Unknown City
df["City"].fillna("Unknown City",inplace=True)
```

```
#Draw a frequency plot for the complaints in each city
df['City'].value_counts().plot(kind="bar",figsize=(12,5))
plt.show()
```

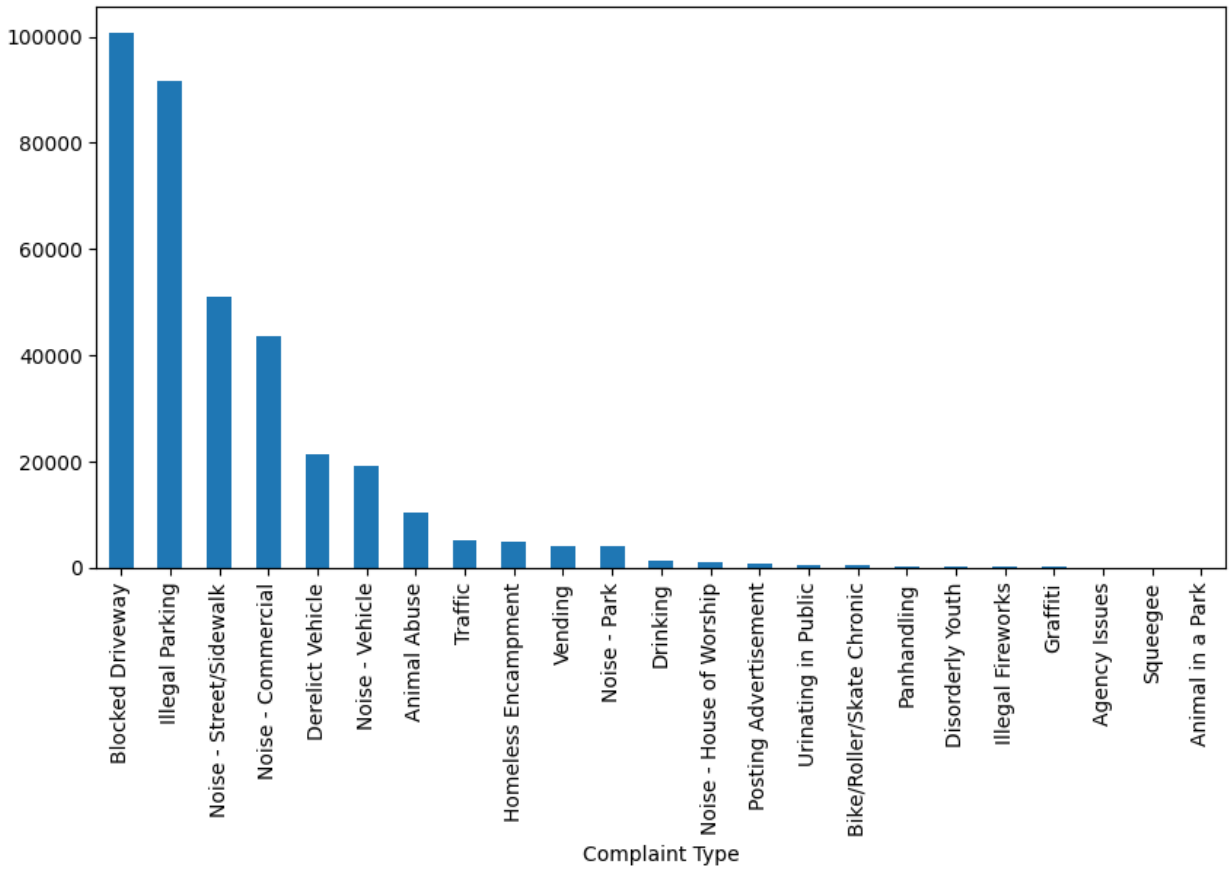



```
#Create a scatter and hexbin plot of the concentration of complaints
across Brooklyn

brooklyn = df[df['City'] == 'BROOKLYN']

#SCATTER
plt.scatter(brooklyn['Longitude'],brooklyn['Latitude'])
plt.show()

#HEXBIN
brooklyn.plot.hexbin(x='Longitude',y='Latitude',gridsize=25)
plt.show()
```


40.75
40.70
40.65
40.60
−74.050 −74.025 −74.000 −73.975 −73.950 −73.925 −73.900 −73.875 −73.850


40.75
40.70
40.65
40.60
Latitude
−74.050 −74.025 −74.000 −73.975 −73.950 −73.925 −73.900 −73.875 −73.850
Longitude
1600
1400
1200
1000
800
600
400
200
0

```
#Plot a bar graph to show the types of complaints
df['Complaint Type'].value_counts().plot(kind='bar',figsize = (10,5))
plt.show()
```




```
#Check the frequency of various types of complaints for New York City
NYC = df[df['City'] == 'NEW YORK']
NYC['Complaint Type'].value_counts()
```

```
Complaint Type
Noise - Street/Sidewalk      22245
Noise - Commercial           18686
Illegal Parking              14549
Noise - Vehicle               6294
Homeless Encampment           3060
Blocked Driveway              2705
Vending                       2638
Animal Abuse                  1941
Traffic                       1769
Noise - Park                  1243
Derelict Vehicle               695
Drinking                       321
Urinating in Public            264
```

```
Bike/Roller/Skate Chronic      254
Noise - House of Worship       222
Panhandling                    206
Disorderly Youth                81
Posting Advertisement           49
Illegal Fireworks               38
Graffiti                        25
Squeegee                         4
Name: count, dtype: int64
```

```
#Find the top 10 complaint types
df['Complaint Type'].value_counts().head(10)
```

```
Complaint Type
Blocked Driveway           100624
Illegal Parking             91716
Noise - Street/Sidewalk     51139
Noise - Commercial          43751
Derelict Vehicle            21518
Noise - Vehicle             19301
Animal Abuse                10530
Traffic                      5196
Homeless Encampment          4879
Vending                      4185
Name: count, dtype: int64
```

```
#Display the various types of complaints in each city
pd.crosstab(df['City'],df['Complaint Type'])
```

```
Complaint Type       Agency Issues  Animal Abuse  Animal in a Park  \
City
ARVERNE                          0            46                 0
ASTORIA                          0           170                 0
Astoria                          0             0                 0
BAYSIDE                          0            53                 0
BELLEROSE                        0            15                 0
BREEZY POINT                     0             2                 0
BRONX                            0          1971                 0
BROOKLYN                         0          3191                 0
CAMBRIA HEIGHTS                  0            15                 0
CENTRAL PARK                     0             0                 0
COLLEGE POINT                    0            35                 0
CORONA                           0           104                 0
EAST ELMHURST                    0            85                 0
ELMHURST                         0            59                 0
East Elmhurst                    0             0                 0
FAR ROCKAWAY                     0           111                 0
FLORAL PARK                      0             7                 0
FLUSHING                         0           191                 0
FOREST HILLS                     0            78                 0
```

FRESH MEADOWS	0	66	0
GLEN OAKS	0	5	0
HOLLIS	0	39	0
HOWARD BEACH	0	51	0
Howard Beach	0	0	0
JACKSON HEIGHTS	0	50	0
JAMAICA	0	317	0
KEW GARDENS	0	26	0
LITTLE NECK	0	21	0
LONG ISLAND CITY	0	40	0
Long Island City	0	0	0
MASPETH	0	56	0
MIDDLE VILLAGE	0	36	0
NEW HYDE PARK	0	1	0
NEW YORK	0	1941	0
OAKLAND GARDENS	0	29	0
OZONE PARK	0	72	0
QUEENS	0	1	1
QUEENS VILLAGE	0	90	0
REGO PARK	0	33	0
RICHMOND HILL	0	55	0
RIDGEWOOD	0	154	0
ROCKAWAY PARK	0	33	0
ROSEDALE	0	44	0
SAINT ALBANS	0	43	0
SOUTH OZONE PARK	0	74	0
SOUTH RICHMOND HILL	0	40	0
SPRINGFIELD GARDENS	0	42	0
STATEN ISLAND	0	786	0
SUNNYSIDE	0	40	0
Unknown City	8	1	0
WHITESTONE	0	43	0
WOODHAVEN	0	57	0
WOODSIDE	0	111	0
Woodside	0	0	0

Complaint Type	Bike/Roller/Skate Chronic	Blocked Driveway	\
City			
ARVERNE	0	50	
ASTORIA	16	3436	
Astoria	0	159	
BAYSIDE	0	514	
BELLEROSE	1	138	
BREEZY POINT	0	3	
BRONX	22	17062	
BROOKLYN	124	36445	
CAMBRIA HEIGHTS	0	177	
CENTRAL PARK	0	0	
COLLEGE POINT	0	597	

CORONA	0	3597
EAST ELMHURST	1	1925
ELMHURST	2	1992
East Elmhurst	0	0
FAR ROCKAWAY	0	383
FLORAL PARK	0	33
FLUSHING	3	3640
FOREST HILLS	6	873
FRESH MEADOWS	0	682
GLEN OAKS	0	48
HOLLIS	0	442
HOWARD BEACH	1	215
Howard Beach	0	1
JACKSON HEIGHTS	2	703
JAMAICA	3	3620
KEW GARDENS	0	429
LITTLE NECK	0	174
LONG ISLAND CITY	3	1052
Long Island City	0	55
MASPETH	1	1000
MIDDLE VILLAGE	1	663
NEW HYDE PARK	0	76
NEW YORK	254	2705
OAKLAND GARDENS	2	177
OZONE PARK	1	1681
QUEENS	0	3
QUEENS VILLAGE	0	772
REGO PARK	0	780
RICHMOND HILL	0	1099
RIDGEWOOD	3	2161
ROCKAWAY PARK	0	80
ROSEDALE	2	270
SAINT ALBANS	0	318
SOUTH OZONE PARK	1	1202
SOUTH RICHMOND HILL	1	1946
SPRINGFIELD GARDENS	0	330
STATEN ISLAND	10	2845
SUNNYSIDE	2	278
Unknown City	2	86
WHITESTONE	4	279
WOODHAVEN	2	1363
WOODSIDE	5	2038
Woodside	0	27

Complaint Type	Derelict Vehicle	Disorderly Youth	Drinking Graffiti \
City			
ARVERNE	32	2	1

```
1
ASTORIA                              426                  5         43
4
Astoria                               14                  0          0
0
BAYSIDE                              231                  2          1
3
BELLEROSE                            120                  2          1
0
BREEZY POINT                           3                  0          1
0
BRONX                               2402                 66        206
15
BROOKLYN                            6257                 79        291
60
CAMBRIA HEIGHTS                      148                  0          0
0
CENTRAL PARK                           0                  0          0
0
COLLEGE POINT                        223                  1          1
2
CORONA                                72                  6         34
4
EAST ELMHURST                        136                  1          9
3
ELMHURST                              94                  2         13
1
East Elmhurst                          2                  0          0
0
FAR ROCKAWAY                         215                  1          4
0
FLORAL PARK                           74                  1          1
0
FLUSHING                             532                  2         47
6
FOREST HILLS                          71                  1          1
3
FRESH MEADOWS                        347                  0          2
0
GLEN OAKS                             57                  0          0
0
HOLLIS                               162                  1          3
0
HOWARD BEACH                         172                  1          4
0
Howard Beach                           0                  0          0
0
JACKSON HEIGHTS                       41                  0         10
1
```

JAMAICA	1132	9	40	3
KEW GARDENS	16	0	1	0
LITTLE NECK	73	2	1	0
LONG ISLAND CITY	220	2	8	3
Long Island City	4	0	0	0
MASPETH	510	2	9	1
MIDDLE VILLAGE	366	0	2	0
NEW HYDE PARK	14	0	0	0
NEW YORK	695	81	321	25
OAKLAND GARDENS	117	1	2	0
OZONE PARK	479	4	20	0
QUEENS	2	0	0	0
QUEENS VILLAGE	478	0	5	1
REGO PARK	94	0	4	1
RICHMOND HILL	200	0	10	1
RIDGEWOOD	507	3	10	3
ROCKAWAY PARK	19	4	23	0
ROSEDALE	247	0	2	2
SAINT ALBANS	248	1	3	0
SOUTH OZONE PARK	425	2	14	2
SOUTH RICHMOND HILL	356	2	25	0
SPRINGFIELD GARDENS	267	0	6	0
STATEN ISLAND	2184	25	188	6
SUNNYSIDE	17	2	12	1
Unknown City	63	0	3	

```
0
WHITESTONE                          279                 1         3
1
WOODHAVEN                           369                 0         4
0
WOODSIDE                            298                 1        15
4
Woodside                              8                 0         0
0

Complaint Type       Homeless Encampment  ...  Noise - House of
Worship  \
City                                      ...

ARVERNE                                 4  ...
14
ASTORIA                                32  ...
21
Astoria                                 0  ...
0
BAYSIDE                                 2  ...
3
BELLEROSE                               1  ...
1
BREEZY POINT                            0  ...
0
BRONX                                 275  ...
90
BROOKLYN                              948  ...
389
CAMBRIA HEIGHTS                         6  ...
2
CENTRAL PARK                            0  ...
0
COLLEGE POINT                           3  ...
2
CORONA                                 26  ...
3
EAST ELMHURST                           2  ...
25
ELMHURST                               34  ...
6
East Elmhurst                           0  ...
0
FAR ROCKAWAY                           16  ...
1
FLORAL PARK                             0  ...
0
FLUSHING                               26  ...
```

```
5
FOREST HILLS                            18  ...
1
FRESH MEADOWS                            6  ...
0
GLEN OAKS                                0  ...
0
HOLLIS                                   9  ...
215
HOWARD BEACH                             3  ...
1
Howard Beach                             0  ...
0
JACKSON HEIGHTS                         11  ...
2
JAMAICA                                 93  ...
15
KEW GARDENS                              5  ...
1
LITTLE NECK                              0  ...
0
LONG ISLAND CITY                        10  ...
0
Long Island City                         0  ...
0
MASPETH                                 11  ...
2
MIDDLE VILLAGE                           5  ...
0
NEW HYDE PARK                            0  ...
0
NEW YORK                              3060  ...
222
OAKLAND GARDENS                          1  ...
0
OZONE PARK                               8  ...
4
QUEENS                                   2  ...
1
QUEENS VILLAGE                          19  ...
2
REGO PARK                                6  ...
1
RICHMOND HILL                           30  ...
0
RIDGEWOOD                               26  ...
2
ROCKAWAY PARK                            4  ...
0
```

```
ROSEDALE                                  4  ...
2
SAINT ALBANS                             11  ...
1
SOUTH OZONE PARK                          5  ...
5
SOUTH RICHMOND HILL                      12  ...
3
SPRINGFIELD GARDENS                       7  ...
1
STATEN ISLAND                            77  ...
18
SUNNYSIDE                                12  ...
0
Unknown City                              1  ...
0
WHITESTONE                                0  ...
0
WOODHAVEN                                10  ...
3
WOODSIDE                                 38  ...
4
Woodside                                  0  ...
0

Complaint Type       Noise - Park  Noise - Street/Sidewalk  Noise -
Vehicle  \
City

ARVERNE                           2                       29
10
ASTORIA                          64                      409
236
Astoria                           0                      145
0
BAYSIDE                           4                       17
24
BELLEROSE                         1                       13
11
BREEZY POINT                      0                        1
1
BRONX                           548                     9144
3556
BROOKLYN                       1575                    13982
5965
CAMBRIA HEIGHTS                   0                       29
100
CENTRAL PARK                      0                      105
0
```

COLLEGE POINT	2	34	140
CORONA	24	243	110
EAST ELMHURST	5	110	82
ELMHURST	35	228	69
East Elmhurst	0	0	0
FAR ROCKAWAY	23	140	83
FLORAL PARK	0	3	2
FLUSHING	61	241	147
FOREST HILLS	20	102	70
FRESH MEADOWS	8	48	97
GLEN OAKS	38	6	4
HOLLIS	17	43	52
HOWARD BEACH	2	22	10
Howard Beach	0	0	0
JACKSON HEIGHTS	8	238	75
JAMAICA	38	365	337
KEW GARDENS	0	13	23
LITTLE NECK	2	10	8
LONG ISLAND CITY	55	133	124
Long Island City	0	28	0
MASPETH	3	124	26
MIDDLE VILLAGE	4	38	45
NEW HYDE PARK	0	0	2
NEW YORK	1243	22245	6294
OAKLAND GARDENS	14	20	

```
7
OZONE PARK                      20                       140
81
QUEENS                           0                         6
2
QUEENS VILLAGE                   2                        69
54
REGO PARK                       22                        64
60
RICHMOND HILL                    4                        93
69
RIDGEWOOD                       28                       448
249
ROCKAWAY PARK                    2                       218
29
ROSEDALE                        69                        26
25
SAINT ALBANS                     1                        81
50
SOUTH OZONE PARK                 4                       108
97
SOUTH RICHMOND HILL              2                        93
93
SPRINGFIELD GARDENS              1                        42
48
STATEN ISLAND                   67                       885
424
SUNNYSIDE                       16                        69
53
Unknown City                     7                        99
9
WHITESTONE                       7                        35
31
WOODHAVEN                        3                        89
81
WOODSIDE                        38                       261
136
Woodside                         0                         5
0

Complaint Type       Panhandling  Posting Advertisement  Squeegee
Traffic  \
City

ARVERNE                         1                       0         0
1
ASTORIA                         2                       3         0
60
Astoria                         0                       0         0
```

0
BAYSIDE 0 0 0
9
BELLEROSE 1 1 0
9
BREEZY POINT 0 0 0
0
BRONX 20 18 0
427
BROOKLYN 49 58 0
1258
CAMBRIA HEIGHTS 0 0 0
7
CENTRAL PARK 0 0 0
0
COLLEGE POINT 0 0 0
16
CORONA 1 1 0
14
EAST ELMHURST 0 1 0
24
ELMHURST 3 1 0
18
East Elmhurst 0 0 0
0
FAR ROCKAWAY 0 0 0
11
FLORAL PARK 0 0 0
0
FLUSHING 2 1 0
59
FOREST HILLS 6 4 0
65
FRESH MEADOWS 1 0 0
15
GLEN OAKS 0 0 0
3
HOLLIS 0 0 0
11
HOWARD BEACH 2 0 0
9
Howard Beach 0 0 0
0
JACKSON HEIGHTS 1 1 0
13
JAMAICA 3 8 0
632
KEW GARDENS 0 1 0
10
LITTLE NECK 0 1 0

```
20
LONG ISLAND CITY                2                       2          0
83
Long Island City                0                       0          0
0
MASPETH                         0                       0          0
71
MIDDLE VILLAGE                  0                       0          0
14
NEW HYDE PARK                   0                       0          0
0
NEW YORK                      206                      49          4
1769
OAKLAND GARDENS                 0                       0          0
6
OZONE PARK                      7                       3          0
21
QUEENS                          0                       0          0
2
QUEENS VILLAGE                  1                       1          0
27
REGO PARK                       0                       0          0
16
RICHMOND HILL                   0                       2          0
8
RIDGEWOOD                       0                       1          0
50
ROCKAWAY PARK                   0                       0          0
7
ROSEDALE                        0                       0          0
25
SAINT ALBANS                    0                       0          0
14
SOUTH OZONE PARK                0                       1          0
36
SOUTH RICHMOND HILL             0                       0          0
12
SPRINGFIELD GARDENS             2                       2          0
12
STATEN ISLAND                  13                     516          0
229
SUNNYSIDE                       0                       3          0
17
Unknown City                    1                       0          0
2
WHITESTONE                      0                       0          0
32
WOODHAVEN                       1                       0          0
7
```

```
WOODSIDE                          0                       0          0
45
Woodside                          0                       0          0
0

Complaint Type       Urinating in Public  Vending
City
ARVERNE                                1        1
ASTORIA                               10       57
Astoria                                0        0
BAYSIDE                                0        2
BELLEROSE                              1        0
BREEZY POINT                           0        0
BRONX                                 54      433
BROOKLYN                             155      575
CAMBRIA HEIGHTS                        0        0
CENTRAL PARK                           0        0
COLLEGE POINT                          0        1
CORONA                                 7       65
EAST ELMHURST                          6        9
ELMHURST                              10       25
East Elmhurst                          0        0
FAR ROCKAWAY                           1       10
FLORAL PARK                            0        0
FLUSHING                              12       37
FOREST HILLS                           2       10
FRESH MEADOWS                          1        1
GLEN OAKS                              2       19
HOLLIS                                 2        0
HOWARD BEACH                           0        5
Howard Beach                           0        0
JACKSON HEIGHTS                        3       86
JAMAICA                               37       24
KEW GARDENS                            3        1
LITTLE NECK                            1        0
LONG ISLAND CITY                       3       31
Long Island City                       0        0
MASPETH                                2        7
MIDDLE VILLAGE                         0        0
NEW HYDE PARK                          0        0
NEW YORK                             264     2638
OAKLAND GARDENS                        0        2
OZONE PARK                             4        1
QUEENS                                 1        0
QUEENS VILLAGE                         5        2
REGO PARK                              1        3
RICHMOND HILL                          5       15
RIDGEWOOD                              9        9
ROCKAWAY PARK                          1        2
```

```
ROSEDALE                                0       19
SAINT ALBANS                            1        2
SOUTH OZONE PARK                        2        5
SOUTH RICHMOND HILL                     1       24
SPRINGFIELD GARDENS                     3        1
STATEN ISLAND                          19       25
SUNNYSIDE                               2       15
Unknown City                            0        1
WHITESTONE                              0        1
WOODHAVEN                               2        6
WOODSIDE                                8       15
Woodside                                0        0

[54 rows x 23 columns]
```

```python
#Create a DataFrame, df_new, which contains cities as columns and complaint types in rows
df_new = pd.crosstab(df['Complaint Type'],df['City'])
df_new
```

```
City                        ARVERNE  ASTORIA  Astoria  BAYSIDE
BELLEROSE  \
Complaint Type

Agency Issues                     0        0        0        0
0
Animal Abuse                     46      170        0       53
15
Animal in a Park                  0        0        0        0
0
Bike/Roller/Skate Chronic         0       16        0        0
1
Blocked Driveway                 50     3436      159      514
138
Derelict Vehicle                 32      426       14      231
120
Disorderly Youth                  2        5        0        2
2
Drinking                          1       43        0        1
1
Graffiti                          1        4        0        3
0
Homeless Encampment               4       32        0        2
1
Illegal Fireworks                 0        4        0        0
1
Illegal Parking                  62     1340      277      638
132
Noise - Commercial                2     1653      310       47
38
```

```
Noise - House of Worship        14       21        0        3
1
Noise - Park                     2       64        0        4
1
Noise - Street/Sidewalk         29      409      145       17
13
Noise - Vehicle                 10      236        0       24
11
Panhandling                      1        2        0        0
1
Posting Advertisement            0        3        0        0
1
Squeegee                         0        0        0        0
0
Traffic                          1       60        0        9
9
Urinating in Public              1       10        0        0
1
Vending                          1       57        0        2
0

City                       BREEZY POINT  BRONX  BROOKLYN  CAMBRIA
HEIGHTS  \
Complaint Type

Agency Issues                          0      0         0
0
Animal Abuse                           2   1971      3191
15
Animal in a Park                       0      0         0
0
Bike/Roller/Skate Chronic              0     22       124
0
Blocked Driveway                       3  17062     36445
177
Derelict Vehicle                       3   2402      6257
148
Disorderly Youth                       0     66        79
0
Drinking                               1    206       291
0
Graffiti                               0     15        60
0
Homeless Encampment                    0    275       948
6
Illegal Fireworks                      0     24        61
1
Illegal Parking                       16   9889     33532
113
```

```
Noise - Commercial                    4   2944     13855
19
Noise - House of Worship              0     90       389
2
Noise - Park                          0    548      1575
0
Noise - Street/Sidewalk               1   9144     13982
29
Noise - Vehicle                       1   3556      5965
100
Panhandling                           0     20        49
0
Posting Advertisement                 0     18        58
0
Squeegee                              0      0         0
0
Traffic                               0    427      1258
7
Urinating in Public                   0     54       155
0
Vending                               0    433       575
0

City                        CENTRAL PARK  ...  SOUTH OZONE PARK  \
Complaint Type                            ...
Agency Issues                          0  ...                 0
Animal Abuse                           0  ...                74
Animal in a Park                       0  ...                 0
Bike/Roller/Skate Chronic              0  ...                 1
Blocked Driveway                       0  ...              1202
Derelict Vehicle                       0  ...               425
Disorderly Youth                       0  ...                 2
Drinking                               0  ...                14
Graffiti                               0  ...                 2
Homeless Encampment                    0  ...                 5
Illegal Fireworks                      0  ...                 1
Illegal Parking                        5  ...               602
Noise - Commercial                     0  ...                82
Noise - House of Worship               0  ...                 5
Noise - Park                           0  ...                 4
Noise - Street/Sidewalk              105  ...               108
Noise - Vehicle                        0  ...                97
Panhandling                            0  ...                 0
Posting Advertisement                  0  ...                 1
Squeegee                               0  ...                 0
Traffic                                0  ...                36
Urinating in Public                    0  ...                 2
Vending                                0  ...                 5
```

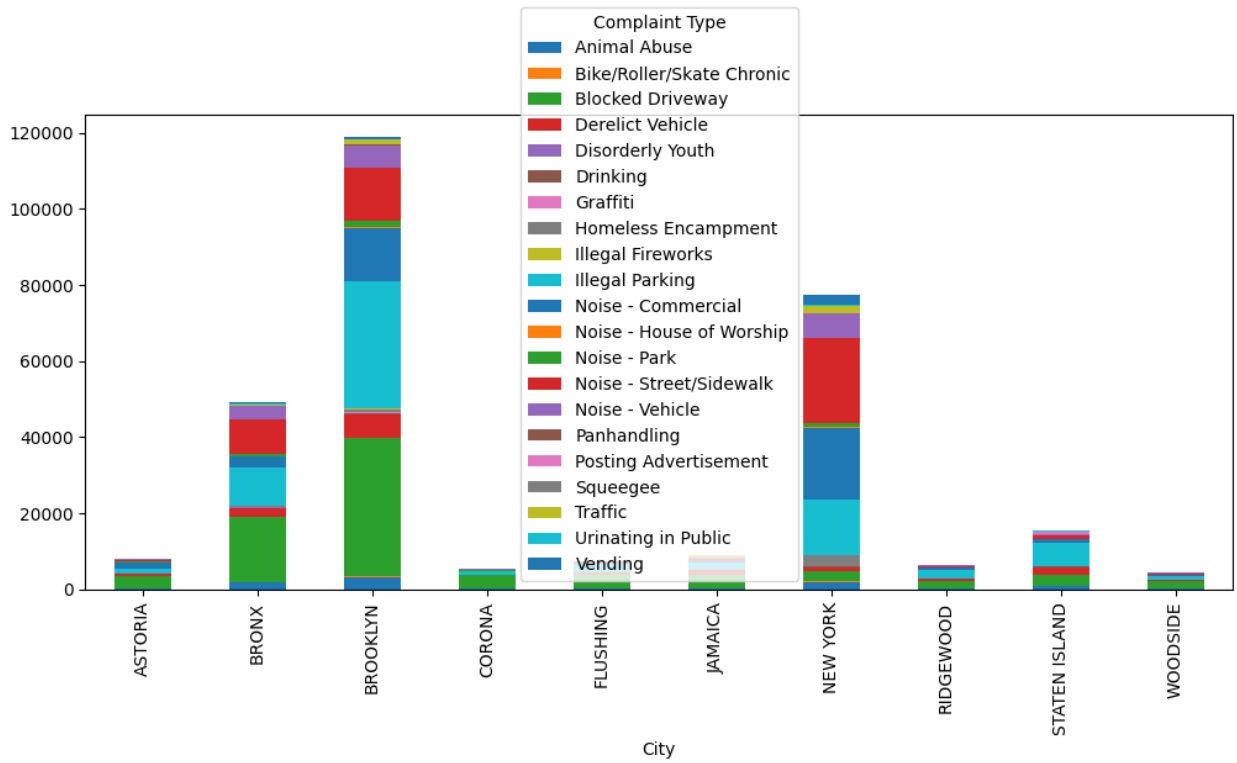
City	SOUTH RICHMOND HILL	SPRINGFIELD GARDENS	\
Complaint Type			
Agency Issues	0	0	
Animal Abuse	40	42	
Animal in a Park	0	0	
Bike/Roller/Skate Chronic	1	0	
Blocked Driveway	1946	330	
Derelict Vehicle	356	267	
Disorderly Youth	2	0	
Drinking	25	6	
Graffiti	0	0	
Homeless Encampment	12	7	
Illegal Fireworks	2	1	
Illegal Parking	596	291	
Noise - Commercial	223	38	
Noise - House of Worship	3	1	
Noise - Park	2	1	
Noise - Street/Sidewalk	93	42	
Noise - Vehicle	93	48	
Panhandling	0	2	
Posting Advertisement	0	2	
Squeegee	0	0	
Traffic	12	12	
Urinating in Public	1	3	
Vending	24	1	

City	STATEN ISLAND	SUNNYSIDE	Unknown City	WHITESTONE	\
Complaint Type					
Agency Issues	0	0	8	0	
Animal Abuse	786	40	1	43	
Animal in a Park	0	0	0	0	
Bike/Roller/Skate Chronic	10	2	2	4	
Blocked Driveway	2845	278	86	279	
Derelict Vehicle	2184	17	63	279	
Disorderly Youth	25	2	0	1	
Drinking	188	12	3	3	
Graffiti	6	1	0	1	
Homeless Encampment	77	12	1		

```
0
Illegal Fireworks                      11          0              0
1
Illegal Parking                      6224        167            312
631
Noise - Commercial                    783        238             79
21
Noise - House of Worship               18          0              0
0
Noise - Park                           67         16              7
7
Noise - Street/Sidewalk               885         69             99
35
Noise - Vehicle                       424         53              9
31
Panhandling                            13          0              1
0
Posting Advertisement                 516          3              0
0
Squeegee                                0          0              0
0
Traffic                               229         17              2
32
Urinating in Public                    19          2              0
0
Vending                                25         15              1
1

City                        WOODHAVEN  WOODSIDE  Woodside
Complaint Type
Agency Issues                       0         0         0
Animal Abuse                       57       111         0
Animal in a Park                    0         0         0
Bike/Roller/Skate Chronic           2         5         0
Blocked Driveway                 1363      2038        27
Derelict Vehicle                  369       298         8
Disorderly Youth                    0         1         0
Drinking                            4        15         0
Graffiti                            0         4         0
Homeless Encampment                10        38         0
Illegal Fireworks                   0         1         0
Illegal Parking                   896      1083       124
Noise - Commercial                209       256         2
Noise - House of Worship            3         4         0
Noise - Park                        3        38         0
Noise - Street/Sidewalk            89       261         5
Noise - Vehicle                    81       136         0
Panhandling                         1         0         0
Posting Advertisement               0         0         0
```

```
Squeegee                                0          0          0
Traffic                                 7         45          0
Urinating in Public                     2          8          0
Vending                                 6         15          0

[23 rows x 54 columns]
```

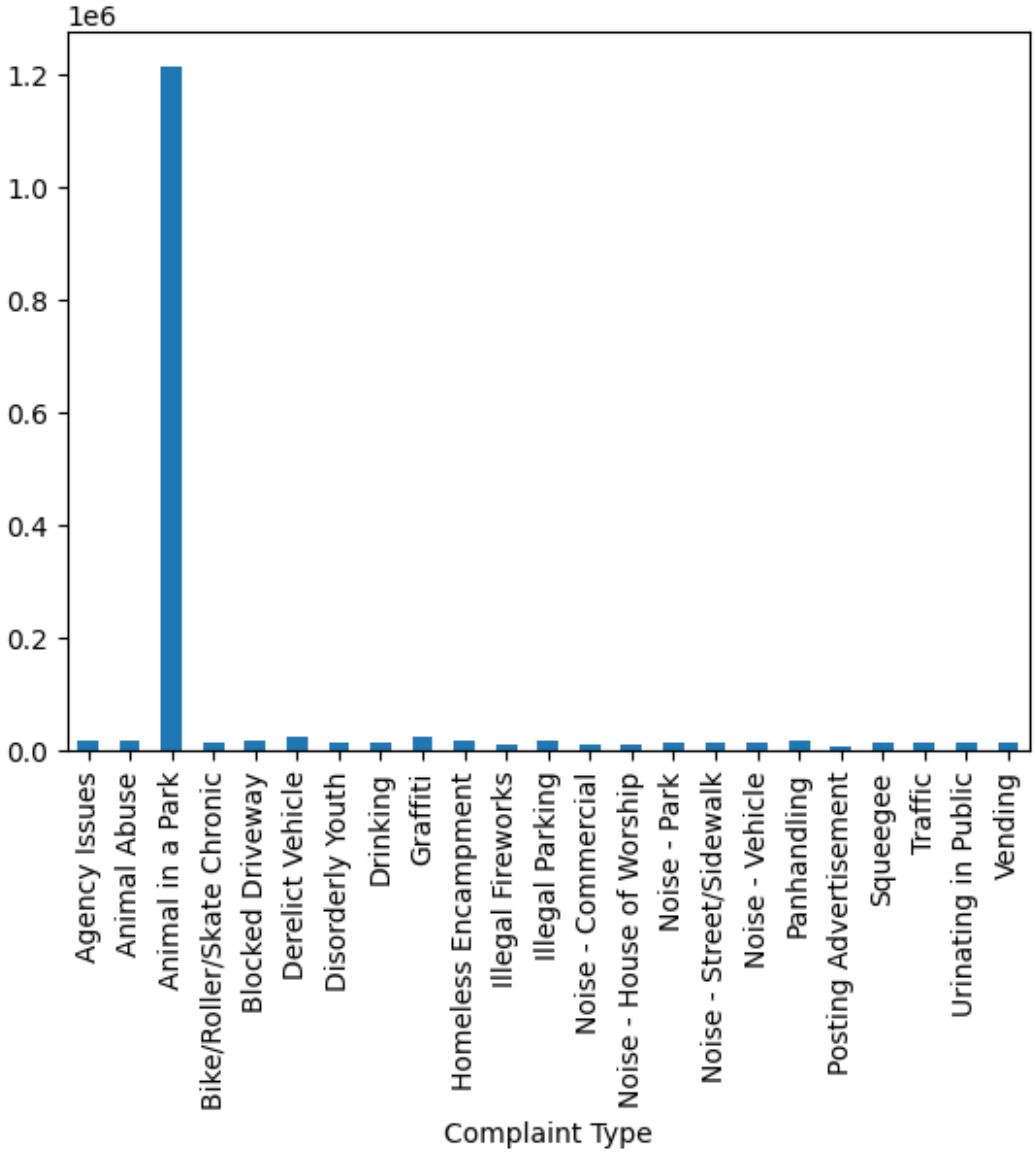
```
#Draw another chart that shows the types of complaints in each city in
a single chart, where different colors show the different types of
complaints

top_cities = df['City'].value_counts().head(10).index
df_filtered = df[df['City'].isin(top_cities)]

df_new = pd.crosstab(df_filtered['City'], df_filtered['Complaint
Type'])
df_new.plot(kind='bar', stacked=True,figsize = (12,5))
plt.show()
```




```
#Sort the complaint types based on the average
Request_Closing_Timegrouping them for different locations

(df.groupby(['City','Complaint Type']))
['Request_Closing_Time'].mean().sort_values().round(2)
```

```
City                   Complaint Type
ARVERNE                Drinking                          859.0
RIDGEWOOD              Posting Advertisement             895.0
EAST ELMHURST          Bike/Roller/Skate Chronic         908.0
OZONE PARK             Illegal Fireworks                1191.0
QUEENS                 Urinating in Public              1226.0
                                                          ...
OAKLAND GARDENS        Homeless Encampment            103156.0
SOUTH RICHMOND HILL    Bike/Roller/Skate Chronic      111277.0
BELLEROSE              Homeless Encampment            140930.0
QUEENS VILLAGE         Graffiti                       191995.0
QUEENS                 Animal in a Park              1212634.0
Name: Request_Closing_Time, Length: 792, dtype: float64
```

```
#Visualize the average of Request_Closing_Time
df.groupby('Complaint Type')
['Request_Closing_Time'].mean().plot(kind='bar')
plt.show()
```




```
!pip install scipy


[notice] A new release of pip is available: 25.3 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip

Collecting scipy
  Downloading scipy-1.17.1-cp314-cp314-win_amd64.whl.metadata (60 kB)
Requirement already satisfied: numpy<2.7,>=1.26.4 in c:\users\surya\
appdata\local\programs\python\python314\lib\site-packages (from scipy)
(2.4.0)
Downloading scipy-1.17.1-cp314-cp314-win_amd64.whl (37.3 MB)
   ---------------------------------------- 0.0/37.3 MB ? eta -:--:--
    ---------------------------------------- 0.8/37.3 MB 6.1 MB/s eta
0:00:06
```

```
   - ---------------------------------------- 1.8/37.3 MB 5.5 MB/s eta
0:00:07
   --- ------------------------------------- 3.1/37.3 MB 5.6 MB/s eta
0:00:07
   ---- ------------------------------------ 4.2/37.3 MB 5.8 MB/s eta
0:00:06
   ----- ----------------------------------- 5.2/37.3 MB 5.5 MB/s eta
0:00:06
   ------ ---------------------------------- 6.3/37.3 MB 5.5 MB/s eta
0:00:06
   ------- --------------------------------- 7.3/37.3 MB 5.4 MB/s eta
0:00:06
   -------- -------------------------------- 8.4/37.3 MB 5.4 MB/s eta
0:00:06
   ---------- ------------------------------ 9.7/37.3 MB 5.4 MB/s eta
0:00:06
   ----------- ----------------------------- 10.7/37.3 MB 5.4 MB/s eta
0:00:05
   ------------ ---------------------------- 12.1/37.3 MB 5.5 MB/s eta
0:00:05
   -------------- ------------------------- 13.4/37.3 MB 5.5 MB/s eta
0:00:05
   --------------- ------------------------ 14.7/37.3 MB 5.6 MB/s eta
0:00:05
   ---------------- ----------------------- 15.7/37.3 MB 5.6 MB/s eta
0:00:04
   ----------------- ---------------------- 16.8/37.3 MB 5.6 MB/s eta
0:00:04
   ------------------- -------------------- 17.8/37.3 MB 5.6 MB/s eta
0:00:04
   -------------------- ------------------- 19.1/37.3 MB 5.6 MB/s eta
0:00:04
   --------------------- ------------------ 20.4/37.3 MB 5.6 MB/s eta
0:00:03
   ------------------------ ---------------- 21.8/37.3 MB 5.7 MB/s eta
0:00:03
   ------------------------- --------------- 23.1/37.3 MB 5.7 MB/s eta
0:00:03
   -------------------------- ------------- 24.4/37.3 MB 5.7 MB/s eta
0:00:03
   ---------------------------- ------------ 25.7/37.3 MB 5.7 MB/s eta
0:00:03
   ----------------------------- ----------- 26.7/37.3 MB 5.7 MB/s eta
0:00:02
   ------------------------------- --------- 28.0/37.3 MB 5.8 MB/s eta
0:00:02
   -------------------------------- -------- 29.4/37.3 MB 5.8 MB/s eta
0:00:02
   --------------------------------- ------- 30.4/37.3 MB 5.7 MB/s eta
```

```
0:00:02
   ---------------------------------- ------ 31.5/37.3 MB 5.7 MB/s eta
0:00:02
   ------------------------------------ ---- 32.8/37.3 MB 5.7 MB/s eta
0:00:01
   ------------------------------------- --- 33.8/37.3 MB 5.7 MB/s eta
0:00:01
   -------------------------------------- -- 35.1/37.3 MB 5.7 MB/s eta
0:00:01
   --------------------------------------- - 36.2/37.3 MB 5.7 MB/s eta
0:00:01
   ----------------------------------------  37.2/37.3 MB 5.7 MB/s eta
0:00:01
   ---------------------------------------- 37.3/37.3 MB 5.7 MB/s
0:00:06
Installing collected packages: scipy
Successfully installed scipy-1.17.1
P-value: 0.0
```

```python
#Perform a Kruskal-WallisH test

from scipy.stats import kruskal

groups = []
for name, group in df.groupby('Complaint Type'):
    groups.append(group['Request_Closing_Time'].values)

stat, p = kruskal(*groups)

print("P-value:", p)
```

```
P-value: 0.0
```

OBSERVATION “The project shows that complaints are concentrated in specific cities and categories, and response times vary significantly across complaint types. Statistical testing confirmed that these differences are meaningful, indicating varying efficiency in handling different issues.”